

Evoking Harmony via Convolution

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Abstract. I show how to evoke the pitch-class content of a chord from an arbitrary source by convolving it with a short impulse response whose partials lie on those pitch-classes in every octave. The preferred Csound implementation (`harmony_convolver_scaled`) builds one joint IR with frequency-dependent grain length, equal-energy normalization across octaves, inverse A-weighting, a dry Dirac, and a single `ftconv`. I contrast the effect with a linear comb and a generic constant- Q bank, and demonstrate it on an unedited twilight field recording beside Château de Lagarde.

Keywords: csound, harmony, convolution, constant- Q , field recording

1 Introduction

As a boy crossing the desert at night I heard melodies and chords in the wind outside the car. This paper reports Csound [1] experiments that evoke such harmonies from varied sources by convolving a signal with an impulse response containing only the pitch-classes of a chosen chord on every octave. Materials are in my GitHub repository [2].

The effect sits at the boundary of timbre and tonality. A constant- Q bank *centered on those pitch-classes* is closely related; a generic constant- Q tiling is only loosely related; a comb filter is less related still (periodic in linear Hz). Gabor/Morlet cells not centered on pitch-classes are a poor fit for this goal.

2 Mathematics

For pitch-class set C ,

$$H_C(f) = \sum_{c \in C} \sum_n H_{c,n}(f), \quad H_{c,n}(f) \simeq H_c(2^{-n}f).$$

With $u = \log_2 f$, octave shift is unit translation and $H_C(u+1) \simeq H_C(u)$: the chord is the unit cell of a log-periodic filter. Grain duration T trades time against frequency resolution [3].

3 Method

`harmony_convolver_scaled` selects MIDI notes whose pitch-classes are in C and assigns each partial f a causal half-cosine of length $T(f) = \min(T, T \cdot 440/f)$. Grains are averaged, a Dirac is added at sample 0, the IR is jointly ℓ_2 -normalized, and `ftconv` [4] is applied once. Full signatures and code are in the repository [2].

Both the causal window and the cap on $T(f)$ serve auditory fusion. Since $Y(f) = H(f)X(f)$ vanishes wherever the source has no energy, the lattice can only expose pitch content the source already contains; what makes that content read as latent in the source rather than as an added voice is keeping the harmonic response inside the source’s own events, since energy persisting after the source falls silent is a cue for a separate resonating object. On an impulse train this design leaves under 3% of each period’s energy outside the interval the dry source occupies, against a third or more for an uncapped or symmetrically windowed grain.

Levelling the octaves takes two stages. Since the sources are broadband, the output power contributed by partial k is $S_0 E_k$ by Parseval’s theorem, so equal presence requires equal grain *energy*, giving $A(f) \propto T(f)^{-1/2}$; a constant factor holds total grain energy fixed so the impulse and Dirac gains keep their meaning. Loudness tracks power rather than spectral density because every grain occupies less than one auditory filter—at most 0.74 of an ERB [5]. Normalizing the window integral instead, which fixes peak transfer gain, tilts the lattice bright by some 8 dB, because the grain bandwidth $\approx 1/T(f)$ widens as the window shortens. The second stage is inverse IEC A-weighting (shelf at 150 Hz, boost capped at +6 dB), which turns equal power into approximately equal loudness; measured band powers follow that curve to within 0.02 dB. Equalizing critical bands rather than partials, which attenuates the crowded low octaves by 4 to 6 dB, is available as an option but is not used here.

A chord-restricted constant- Q bank is a near neighbour (same lattice, ongoing IIR filtering). A generic three-semitone constant- Q tiling lacks chord identity. A comb with delay $1/f_0$ has equal linear-Hz spacing.

4 Results

Figure 1 compares click and noise under dry, comb ($f_0 = 200$ Hz), sparse constant- Q (every three semitones), and evoke with PCs 0, 4, 7, 11 (CM7), loudness-matched to within 1.1 dB A-weighted within each source group. Comb is linear-periodic; constant- Q is log-periodic but not chordal; the convolver shows the CM7 lattice, articulated to the top of the audible range under equal-energy normalization.

A twilight walk beside Château de Lagarde (Tascam DR-40; unedited source) is processed under score control of T , wet/dry, and pitch-classes [6]. Moments include chord colour under voices, bells, “Bonjour,” and harmony in footsteps and jackdaws.

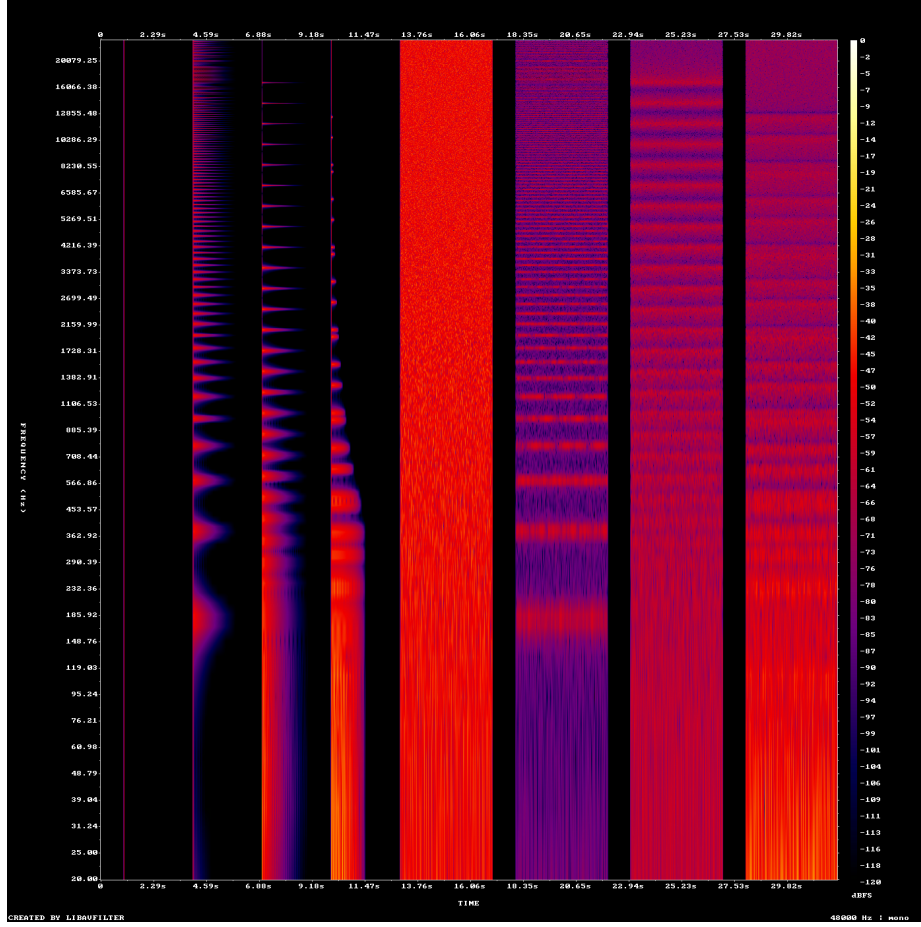


Fig. 1. Comparison spectrogram: click / comb / CQT / evoke CM7, then the same four treatments of noise.

5 Discussion

Evoking harmony is chord-specific and log-periodic, unlike a comb, and uses a finite causal grain plus dry Dirac rather than a steady filterbank. It is most appealing when harmonic sensation remains subtle relative to the source; wetter settings yield a clearer shadow or ring. supplementary figures and audio on GitHub [2] include audio and orchestra code.

Note. This abbreviated version was edited for length with assistance from generative AI tools; the author reviewed the text and retains responsibility for all technical content.

References

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